

$$f(t) = t^2$$

Vom Calcula. gib es noch Ableitungen

$$a = \frac{\int_0^1 [g(x) \cdot (x^2 - x)]}{\int_0^1 [(x^2 - x)^2]} \quad \text{und } g(x) = f(x) - x$$

$$\text{cu } f(x) = x^2$$

$$b = 1 - a$$

$$\begin{aligned} \text{des: } \int_0^1 [(x^2 - x)^2] &= \int_0^1 (x^4 - 2x^3 + x^2) dx = \\ &= \left(\frac{x^5}{5} - 2 \frac{x^4}{4} + \frac{x^3}{3} \right) \Big|_0^1 = \frac{1}{5} - \frac{2}{2} + \frac{1}{3} = \frac{1}{5} - \frac{1}{2} + \frac{1}{3} \end{aligned}$$

$$= \frac{6 - 15 + 10}{30} = \frac{1}{30} \quad (1)$$

$$\int_0^1 [g(x) \cdot (x^2 - x)] = \int_0^1 [(x^2 - x) \cdot (x^2 - x)] dx =$$

$$= \int_0^1 (x^{n+2} - x^{n+1} - x^3 + x^2) = \left(\frac{x^{n+3}}{n+3} - \frac{x^{n+2}}{n+2} - \frac{x^4}{4} + \frac{x^3}{3} \right)$$

$$= \frac{1}{n+3} - \frac{1}{n+2} - \frac{1}{4} + \frac{1}{3} = \frac{1}{n+3} - \frac{1}{n+2} + \frac{1}{12}$$

$$\text{des: } a = \frac{\frac{1}{n+3} - \frac{1}{n+2} + \frac{1}{12}}{\frac{1}{30}} = \frac{30}{n+3} - \frac{30}{n+2} + \frac{5}{2}$$

$$\boxed{a = \frac{30}{n+3} - \frac{30}{n+2} + \frac{5}{2}}$$

$$b = 1 - a.$$

Deci polinomul aproximant al lui f este

$$P(x) = \left(\frac{30}{n+3} - \frac{30}{n+2} + \frac{5}{2} \right) \cdot (x^2 - x) + x$$

$$P(x) = \underbrace{\left(\frac{30}{n+3} - \frac{30}{n+2} + \frac{5}{2} \right)}_a \cdot X^2 + \underbrace{\left(1 - \frac{30}{n+3} + \frac{30}{n+2} - \frac{5}{2} \right)}_b \cdot X$$

Pentru $n=2$

$$P(x) = \left(\frac{30}{5} - \frac{30}{3} + \frac{5}{2} \right) X^2 + \left(1 - \frac{30}{5} + \frac{30}{3} - \frac{5}{2} \right) \cdot X$$

$$P(x) = \left(5 - 6 + \frac{5}{2} \right) x^2 + \left(1 - 5 + 6 - \frac{5}{2} \right) X$$

$$P(x) = \frac{3}{2} X^2 - \frac{1}{2} X.$$

Graficul funcției $f(x) = \frac{3}{2}x^2 - \frac{1}{2}x$ împreună cu graficul funcției $g(x) = x^3$ pot fi vizualizați în ~~un~~ imaginea din directorul curent.